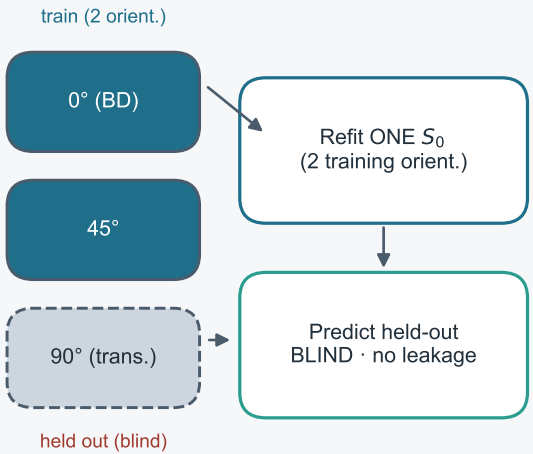
A leakage-free validation protocol and mechanism-keyed applicability domain for orientation-resolved ductile damage descriptors in additively manufactured metals

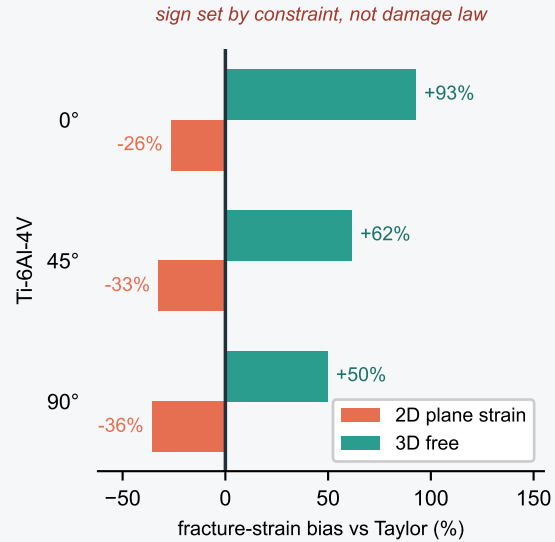
validated out-of-sample · homogenization bias quantified · applicability gated — a screening rule that is never worse than the baseline

A Leakage-free validation



orientation multipliers removed · literature g_s · 9 folds

B Taylor bias flips sign



C Applicability gate

